

# A LEADING ENERGY & INFRASTRUCTURE COMPANY

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**API Industry**

# Pharma/API Industry Snapshot

- Indian pharmaceutical industry (IPI) is ranked **3<sup>rd</sup>** globally in terms of **volume** and **13<sup>th</sup>** in terms of **value**
- **70%** of the industry's revenues comes from **generic medicines**

- API contributes **~35%** of the Indian pharmaceutical market.
- India is the **3<sup>rd</sup>** largest producer of API globally - **500 API** manufacturers contributing **8%** in the global API industry.
- **500+ different APIs** are manufactured in India.

Major Segments of the Pharmaceutical Industry are Generic drugs, OTC Medicines and **API/Bulk Drugs**, Vaccines, **Contract Research & Manufacturing**, Biosimilars & Biologics

## What is API?

The central, biologically **active components** of a drug that produce the intended therapeutic effect. Some medicines contain more than one active pharmaceutical ingredient that act in different ways in the body.

## API vs Normal medicine

- Normal medicine refers to the finished drug product that consumers take.
- Active Pharmaceutical Ingredient is the core chemical component that provides the therapeutic effect.

## API OVERVIEW

## PLI Scheme

- Launched in 2021 with an outlay of Rs **15,000 crore** for 2020-2029.
- Scheme focuses on **41 critical APIs/KSMs**

Under the PLI scheme, manufacturing of **35 APIs**, representing around **67% of APIs** for which India has **90% import dependence** has already started.

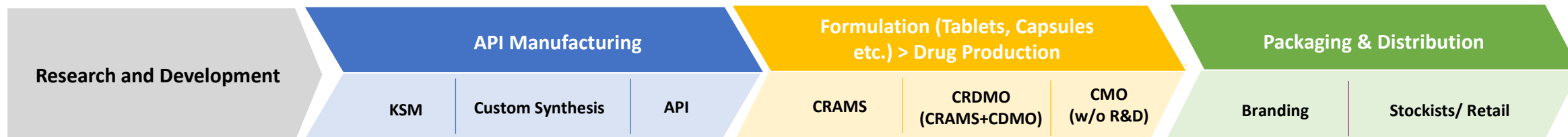
## Technological Advances in API Manufacturing –

Innovations like continuous-flow systems, green chemistry, and AI-driven synthesis enhance efficiency, quality, and scalability.

**Impact on the API Sector** – Lower production costs and waste, and drive investment in hi-tech API facilities.



# Understanding the API/ Pharma Value Chain



## Recent Trends:

- **Low-margin generics to high margin innovative drugs**, biosimilars, and complex injectables
- **Patent expiration** of blockbuster drugs > **launch generic versions**
- Partnerships with **CROs**

## Key Players:



## Importance:

- Helps companies to obtain specialized molecules, intermediates, APIs
- Reduces the **time** and **cost** of drug development
- Supports small-scale and pilot production

## Key Players:



## Snapshot:

- KSM imports come from China
- PLI Scheme is promoting domestic API and KSM production.

## Key Players:



## Importance:

- Clinical trials is **time consuming R&D**, companies and Innovators Outsource it to third party providers
- Outsourcing R&D, manufacturing, and regulatory services
- Partnerships with CRAMS

## Key Players:



## Recent Trends:

- Pharma companies do in-house branding and packaging
- Multiple companies only do packaging and branding
- Now-a-days retailing is gaining traction.

## Key Retail Players:



**Research and Development** : Developing new drugs or improving existing formulations  
(R&D required from API development to final drug production and majority of the players are into R&D)

CRAMS: Contract Research and Manufacturing Services  
CRDMO: Contract Research, Development, and Manufacturing Organization  
CMO: Contract Manufacturing Organizations

# About APIs

>50% APIs are **imported** in India and majority of imports are from **China**

Key APIs for which India is dependent on China are:

- Penicillin G
- Levodopa
- Streptomycin
- Meropenem
- Carbidopa
- Vancomycin
- Gentamicin



## Key Starting Materials (KSMs)

- **Raw materials** or intermediates
- Constitutes a significant **structural fragment** in the structure of API

## Active Pharmaceutical Ingredients (APIs)

- **Primary active ingredient**
- Manufactured in the **initial** stage of drug production

## Drug Formulations

- Combines **inert materials** and **excipients** with APIs
- Produce viable drug products with **desired properties**



## Key starting materials (KSM)

Raw materials for APIs, usually simple chemical substances

## Intermediates

Chemical compounds in the process of producing APIs

Many API manufacturers have to import raw materials; no matter where an API is produced, the materials come mainly from China

## Active pharmaceutical ingredients (API)

Components to treat illness

## Excipient

Substances used in finished drug formula, such as lactose and starch

## Drug products

# Understanding the API Value Chain

## Research and Development

### Drug Discovery:

Identification of potential compounds through screening, synthesis, or biotechnological methods.

**Preclinical Testing:** In vitro and in vivo studies to determine biological activity, toxicity, and pharmacokinetics.

**Process Development :** Designing a scalable and efficient synthetic route for the API (chemical or biological synthesis).

## Raw material sourcing

### Starting Materials:

Procurement of raw chemical or biological materials (intermediates, reagents, solvents).

### Supply Chain

**Management:** Sourcing high-quality materials from certified suppliers globally (e.g., China, India, EU).

**Compliance:** Ensuring materials meet cGMP (Current Good Manufacturing Practices) and pharmacopeial standards

## API Manufacturing

### Chemical Synthesis:

Multi-step synthesis in reactors

### Purification:

Crystallization, chromatography, or filtration to purify the API.

### Quality Control (QC)\*:

Testing for identity, purity, potency, and residual solvents.\*

### GMP Compliance:

Manufacturing must follow regulatory guidelines (FDA, EMA, WHO).

## Formulation (If in-house)

**API to Drug Product:** If integrated, the API is then formulated into tablets, capsules, injectables, etc., with excipients.

### Compatibility Studies:

Ensuring API works well with other formulation ingredients.

## Packaging and Distribution

**API Packaging:** In bulk drums or specialized containers (moisture/light protected).

**Storage:** Controlled environments (temperature, humidity).

**Distribution:** Transport to pharmaceutical formulation plants globally.

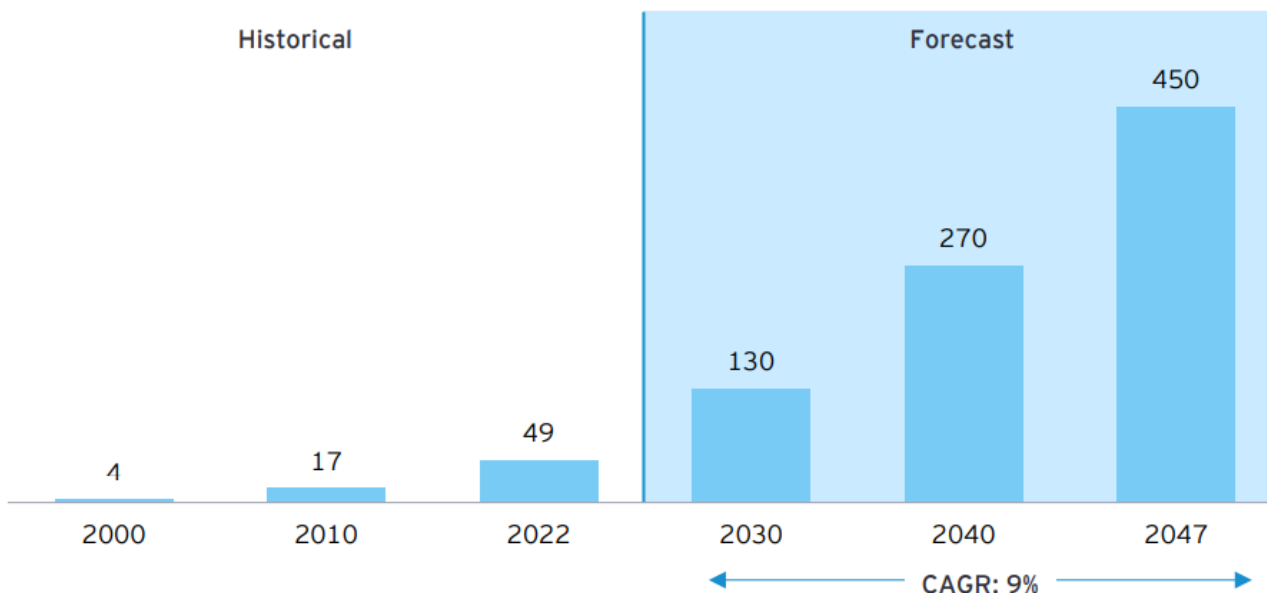


# Market Analysis (Pharma & API)

Indian pharmaceutical industry - US\$b  
(domestic market + exports)

Historical

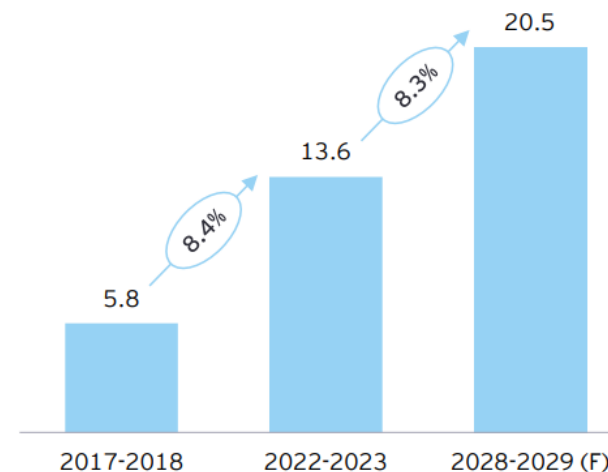
Forecast



- Indian Pharma market valued at approximately **\$55 billion (2024)** and is expected to reach **\$130 billion** by 2030.
- In India, more than **10,000 manufacturing facilities**, over **3,000 pharma companies**, and **650 US-FDA-compliant pharma plants**—the largest number outside the US.
- In 2023, India exported **\$27 billion** (1.5 MMT) and ranked 11<sup>th</sup> in the list of top pharmaceutical exporting nations.

Indian API industry

Indian API industry - market size (US\$b)



## Key initiatives towards self-reliance

### Government:

- PLI scheme for bulk drugs and drug intermediates incentivizing companies to manufacture KSMs/APIs
- Establishment of bulk drug parks

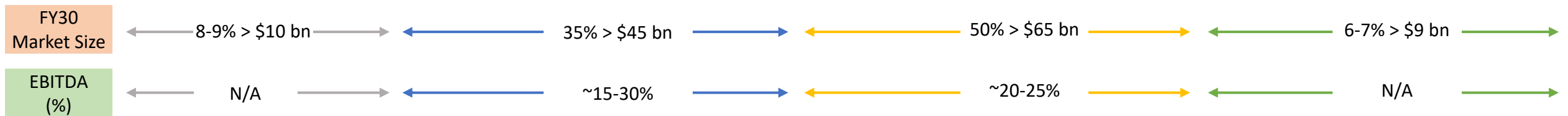
### Academia:

- Actively researching cost-efficient, clean, and scalable synthesis routes to support indigenous production

### Industry:

- Capacity expansion / dedicated subsidiary
- Vertical integration to secure supply chain and maintain cost-competitiveness

- APIs have been identified as one of the 13 pivotal categories under the Atmanirbhar Bharat Abhiyan.
- Under PLI scheme, production of **35 of the 41** selected APIs already underway.
- There has also been a notable decrease in API prices. For instance, the cost of the paracetamol API, which had surged to **INR900 per kg** during the COVID-19 pandemic, has now receded to **INR250 per kg** as of April 2024.



|                                                              |                                                                                     | Focus on API |                                                                                           | Focused Offerings |                                        |                  |                     |   |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------------|-------------------|----------------------------------------|------------------|---------------------|---|
| Companies manufacturing API and Final Drug with in-house R&D |    | ←            | Moderate API manufacturing for in-house consumption; No 3 <sup>rd</sup> Party             | →                 | Formulations+<br>+ Wellness + Vaccines | →                | In-house Branding   | → |
|                                                              |    | ←            | Moderate API manufacturing for in-house consumption; No 3 <sup>rd</sup> Party             | →                 | Formulations + API<br>+ Respiratory    | →                | In-house Branding   | → |
|                                                              |    | ←            | Moderate API manufacturing for in-house consumption; No 3 <sup>rd</sup> Party             | →                 | Formulations + Specialty<br>+ Generics | →                | In-house Branding   | → |
|                                                              |   | ←            | Lower API manufacturing for in-house consumption; No 3 <sup>rd</sup> Party                | →                 | Formulations + Specialty<br>+ Generics | →                | In-house Branding   | → |
|                                                              |  | ←            | Limited API manufacturing for in-house consumption; No 3 <sup>rd</sup> Party              | →                 | Branded Formulations<br>(India Focus)  | →                | In-house Branding   | → |
| Major API Manufacturers with in-house R&D                    |  | ←            | Major API manufacturing for in-house consumption; Supplies to 3 <sup>rd</sup> Party also  | →                 | Formulations<br>+ APIs                 | →                | In-house Branding   | → |
|                                                              |  | ←            | Strong API manufacturing for in-house consumption                                         | →                 | API Core Focus                         | →                | In-house Branding   | → |
|                                                              |  | ←            | Strong API manufacturing for in-house consumption; Supplies to 3 <sup>rd</sup> Party also | →                 | Branded Formulations<br>(India Focus)  | →                | In-house Branding   | → |
|                                                              |                                                                                     |              |                                                                                           |                   | ↔ Focus on Drug Production             | ↔ API Production | ↔ In-House Branding |   |

# Competition



|                      |  |  |  |  |  |  |  |  |  |  |
|----------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Estd. Year           | 1983                                                                              | 1984                                                                              | 1986                                                                              | 1935                                                                               | 1952                                                                                | 1968                                                                                | 1995                                                                                | 1959                                                                                | 1990                                                                                | 1907                                                                                |
| Global Presence      | 100+ countries                                                                    | 60+ countries                                                                     | 150+ countries                                                                    | 80+ countries                                                                      | 50+ countries                                                                       | 100+ countries                                                                      | India, semi-regulated markets                                                       | Select markets (Brazil, Germany, etc.)                                              | 90+ countries                                                                       | 60+ countries                                                                       |
| Market Share (India) | 9-10 %                                                                            | 6%                                                                                | 5-6%                                                                              | ~5%                                                                                | 4%                                                                                  | 4%                                                                                  | ~5%                                                                                 | 2%                                                                                  | <2%                                                                                 | 1%                                                                                  |
| Business Type        | Formulations + Specialty + APIs                                                   | Formulations + APIs + Biologics                                                   | Formulations + APIs                                                               | Formulations + Respiratory + APIs                                                  | Formulations + Biologics + Wellness                                                 | Formulations + Specialty + Generics                                                 | Branded Formulations (India focus)                                                  | Formulations + India branded                                                        | APIs (CRAMS-focused)                                                                | Formulations + APIs                                                                 |
| API Business         | Internal + moderate third-party                                                   | Strong API + third-party                                                          | Major API supplier                                                                | Moderate API business                                                              | Moderate (incl. for vaccines)                                                       | Moderate                                                                            | Limited                                                                             | Low                                                                                 | API Core focus                                                                      | Strong in antibiotics + select APIs                                                 |
| Key Strengths        | Specialty, US generics, dermatology                                               | Complex generics, biosimilars, injectables                                        | APIs, injectables, US filings                                                     | Respiratory, HIV, India + Africa                                                   | Biosimilars, vaccines, wellness (OTC)                                               | US inhalation, women's health                                                       | Strong India brands, DTC sales model                                                | Strong in CV, diabetes, CNS in India                                                | CRAMS, API exports                                                                  | Chronic-focused branded generics                                                    |

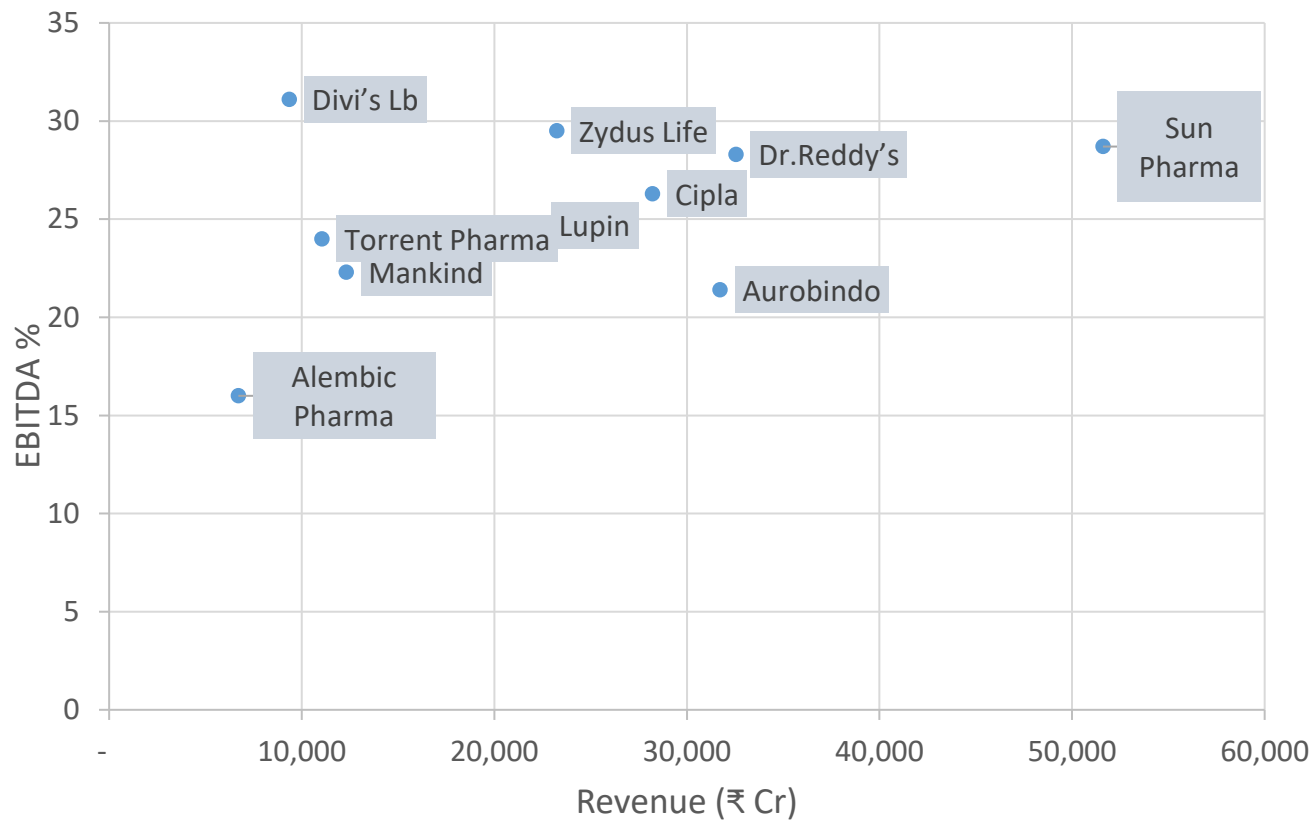
OTC: Over the Counter like Volini, Revitals etc  
 ARV: Anti Retro Viral medicines  
 AMR: Antimicrobial Resistance



# Competition



Revenue vs EBITDA % (FY25)

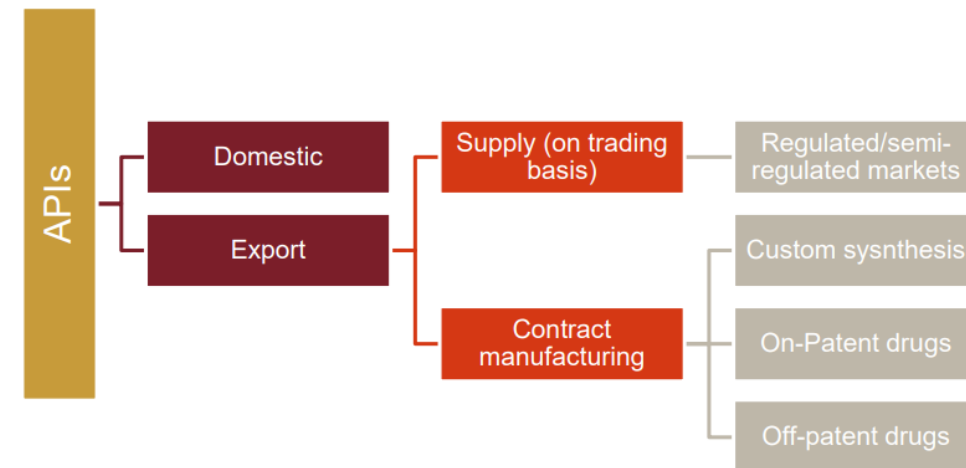


| Category                   | Leaders                                                       |
|----------------------------|---------------------------------------------------------------|
| India Formulations         | Sun Pharma, Mankind, Cipla, Zydus                             |
| US Generics                | Sun Pharma, Aurobindo, Dr. Reddy's                            |
| APIs / CRAMS               | Divi's Labs, Aurobindo, Dr. Reddy's, Alembic                  |
| Biosimilars & Biologics    | Zydus, Dr. Reddy's, Cipla                                     |
| Consumer Health (OTC)      | Zydus Wellness, Cipla Health, Mankind                         |
| Specialty/Complex Generics | Sun Pharma, Dr. Reddy's, Cipla, Zydus                         |
| Innovation/NCEs            | Zydus (Saroglitazar), Sun (Ilumya), Dr. Reddy's (biosimilars) |
| India-Only Powerhouses     | Mankind, Torrent                                              |
| CRAMS / CDMO Leadership    | Divi's Labs                                                   |

# API Focused Companies

| Company                                          | Business Overview                                                                                                                                      | Revenue – FY24 (₹ Cr) | ~EBITDA – FY24 (%) |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------------|
| <b>Solara Active Pharma Sciences Ltd (SAPSL)</b> | <ul style="list-style-type: none"> <li>International API supplier, CRAMS</li> <li>10+ APIs</li> <li>6 API manufacturing facilities in India</li> </ul> | 1,294                 | -4%                |
| <b>Divi's Laboratories</b>                       | <ul style="list-style-type: none"> <li>2 manufacturing facilities in India</li> </ul>                                                                  | 8,002                 | 26.2%              |
| <b>Laurus Labs</b>                               | <ul style="list-style-type: none"> <li>Generics API, Generics FDF and Synthesis</li> <li>11 manufacturing facilities</li> </ul>                        | 1,195                 | 16.5%              |
| <b>IOL Chemicals &amp; Pharmaceuticals (ICP)</b> | <ul style="list-style-type: none"> <li>10 API facilities and 3 specialty chemicals facilities</li> </ul>                                               | 2,163                 | 12.5%              |
| <b>SMS Pharmaceuticals</b>                       | <ul style="list-style-type: none"> <li>4 multi product facilities</li> <li>40+ products, 30+ process patents</li> </ul>                                | 709                   | 15.9%              |
| <b>Neuland Laboratories Ltd (NLL)</b>            | <ul style="list-style-type: none"> <li>100+ APIs across 10 therapeutic areas</li> <li>3 manufacturing plants in India</li> </ul>                       | 1,559                 | 30.1%              |
| <b>Granules India Ltd (GIL)</b>                  | <ul style="list-style-type: none"> <li>7 manufacturing plants</li> </ul>                                                                               | 4,506                 | 18.3%              |
| <b>Hikal Ltd</b>                                 | <ul style="list-style-type: none"> <li>APIs, intermediates</li> <li>CRDMO</li> <li>3 manufacturing facilities in India</li> </ul>                      | 1,100                 | 14.5%              |
| <b>Suven Pharmaceuticals</b>                     | <ul style="list-style-type: none"> <li>APIs and intermediates</li> <li>CRDMO</li> </ul>                                                                | 1,080                 | 42.8%              |

Revenue Model Adopted by bulk drug players



Source: CRISIL Research

# Market Analysis (CRDMO)

## The US BIOSECURE Act (enacted on 9 September 2024)

### What it means: Decoupling from China

- Prohibits US federal agencies and recipients of federal funds from contracting with or procuring services and biotechnology equipment from "Chinese biotechnology companies of concern"
- Will extend to companies that source or utilize equipment or services from the listed companies

Window of  
opportunity  
for India's  
pharma  
industry

### Why is India a contender?

- Opportunity for the growth of India's CRDMO industry as India supplies 40%-50% of all generic drugs in the US
- India houses the most FDA-approved pharmaceutical manufacturing units outside the US
- Highly skilled workforce, relatively low manufacturing cost

### Key focus areas for harnessing opportunity

#### Increase focus on quality

Need to build quality culture and strengthen audit mechanisms

#### Critical to achieve self-sufficiency in APIs

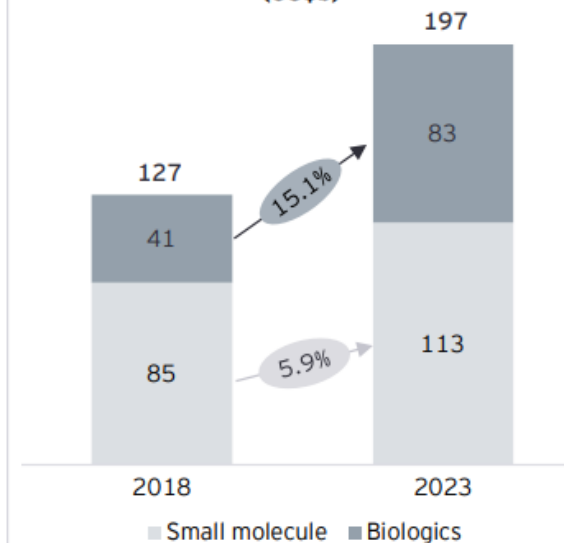
The act would extend to foreign entities associated with "biotechnology companies of concern" suggesting that India's API sourcing from China could be flagged

#### Become cost competitive

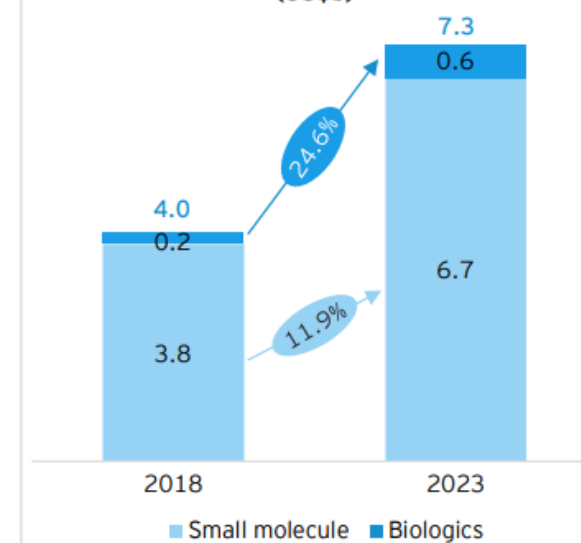
Building scale (bio-manufacturing capacity as well as capability in novel modalities) and efficiencies

CRDMO market is experiencing a paradigm shift towards large molecules, with their share increasing from 35% in 2018 to approximately 42% in 2023

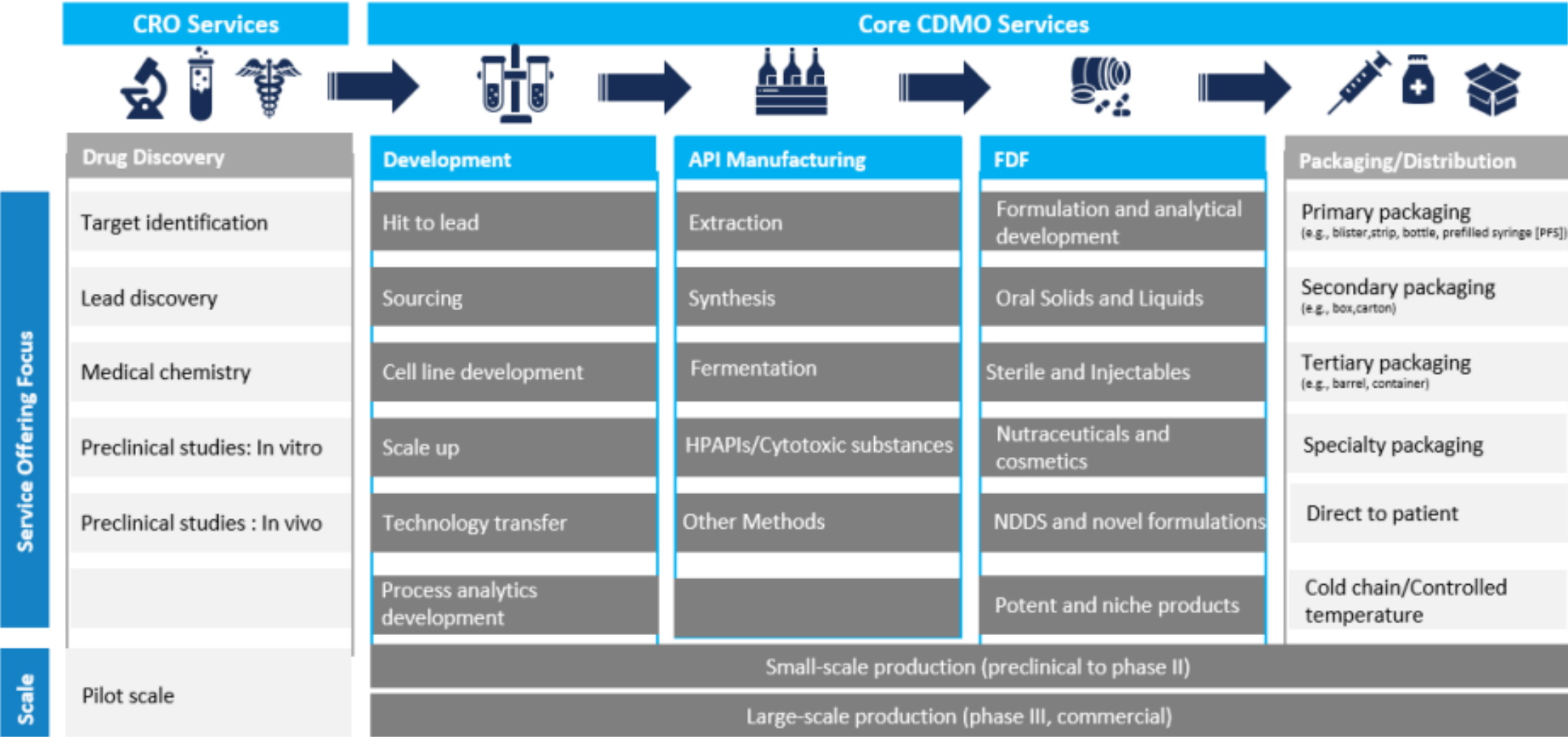
Global CRDMO market size by modality (US\$b)



Indian CRDMO market size by modality (US\$b)



# Service Spectrum - CRDMO

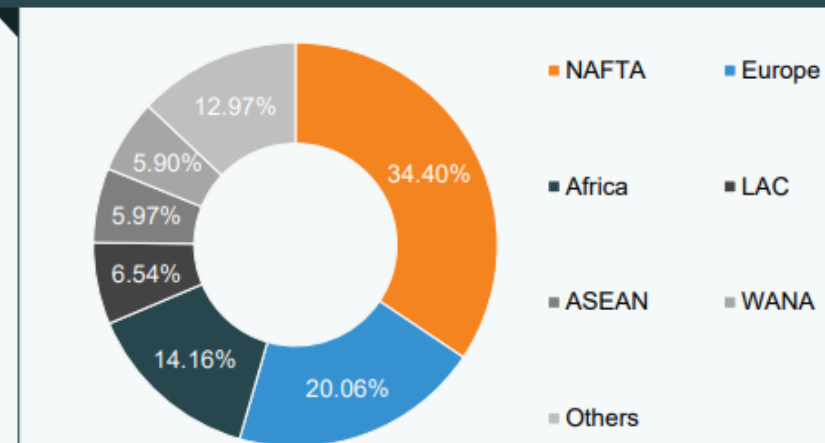




# Market Analysis (Export - Import)

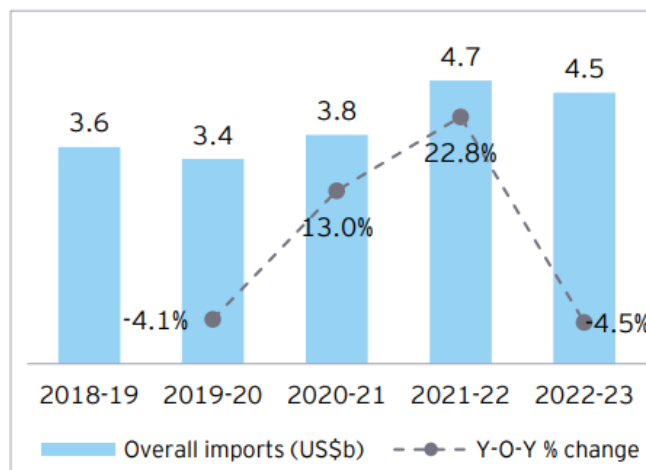
- India's pharma supplies reach approximately **200 countries** and territories worldwide, supplying close to **50% of Africa's requirement** for generics and **40% of US generics** demand.
- ~72% of overall bulk drug/APIs** and intermediates imports by India in FY24 are from **China**, vs. 66% in FY22.
- About **20%** of the global **exports** in **generic drugs** are met by India.

Major Export Destinations in India's Pharma Export in FY24 (%)

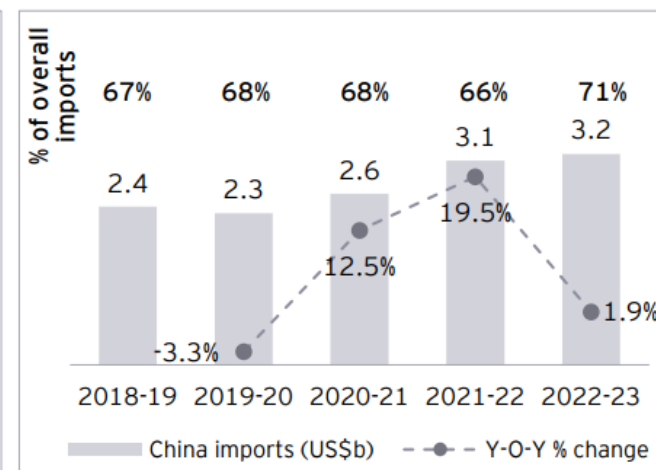


India import dynamics (2018-2023): Bulk drugs (APIs) and intermediates

Total imports (US\$b)



China imports (US\$b)



| Particulars | Direct Imports (FY23) | Developed in India (FY23) | Exports (FY23)  |
|-------------|-----------------------|---------------------------|-----------------|
| KSM         | N/A                   | N/A                       | N/A             |
| API         | ~\$4.5 bn             | ~\$9.1 bn                 | <b>\$4.4 bn</b> |
| Pharma      | \$8.2 bn              | \$47 bn                   | ~\$27 bn        |

**3,44,053.2 MT** out of the total 4,51,894.3 MT of overall pharma raw material imports, from China, in the year 2023-24

## Relative cost comparison in India

|                              |       |
|------------------------------|-------|
| Cost in developed countries  | 100%  |
| Production cost in India     | 50%   |
| R&D cost in India            | 12.5% |
| Clinical trial cost in India | 10%   |

Source: Pharmexcil

Note: Costs in India as % of costs in developed countries

NAFTA: North American Free Trade Agreement  
WANA: West Asia and North Africa  
LAC: Latin America and Caribbean

# API - Growth Drivers in India



## Lower cost of production

India accounts for ~**10% of global** pharma production by volume, driven by low costs—**production is 50%, R&D 12.5%, and clinical trials 10%** of developed countries—making it a key hub for global outsourcing.



## Patent cliff

By 2026, **patents on drugs worth \$240 billion will expire**, allowing generic versions to enter the market. This creates a \$5–6billion opportunity for Indian pharma companies, which are preparing to launch these generics as patents expire.



## Supply Chain disruptions

Geopolitical risks and supply disruptions from China and the Middle East have pushed India to boost domestic API production. This shift toward self-reliance and export diversification is driving long-term growth in the Indian API industry.



**India's rising per capita income** is increasing demand for healthcare, as more people can afford better medical services. From FY13 to FY23, per capita GDP grew at a 4.5% CAGR, except for a 6.8% drop in FY21 due to the Covid-19 lockdown and job losses. However, since income growth is uneven, higher healthcare spending will mainly come from wealthier households.



## Increasing contract manufacturing and outsourcing activities

Rising global demand for pharmaceuticals, combined with limited API production capacity in many countries, makes India a preferred destination for contract manufacturing and outsourcing due to its low costs and expertise.

# Production Linked Incentive (PLI) Scheme

Companies under Target Segment II (Fermentation Based Niche KSMs/Drug Intermediates/APIs) are as follows:

| Name of Approved Applicant          | Committed Production Capacity (in MT) | Committed Investment (in Rs. crore) | Committed Investment (in US\$ million) |
|-------------------------------------|---------------------------------------|-------------------------------------|----------------------------------------|
| Natural Biogenex Private Limited    | 12                                    | 31.43                               | 4.31                                   |
| Natural Biogenex Private Limited    | 10                                    | 26.19                               | 3.59                                   |
| Natural Biogenex Private Limited    | 15                                    | 39.29                               | 5.39                                   |
| Symbiotec Pharmed Private Limited   | 15                                    | 5.00                                | 0.69                                   |
| Macleods Pharmaceutical Limited     | 200                                   | 198.36                              | 27.19                                  |
| Optimus Drugs Private Limited       | 200                                   | 35.00                               | 4.80                                   |
| Sudarshan Pharma Industries Limited | 200                                   | 57.00                               | 7.81                                   |
| Optimus Drugs Private Limited       | 50                                    | 30.00                               | 4.11                                   |

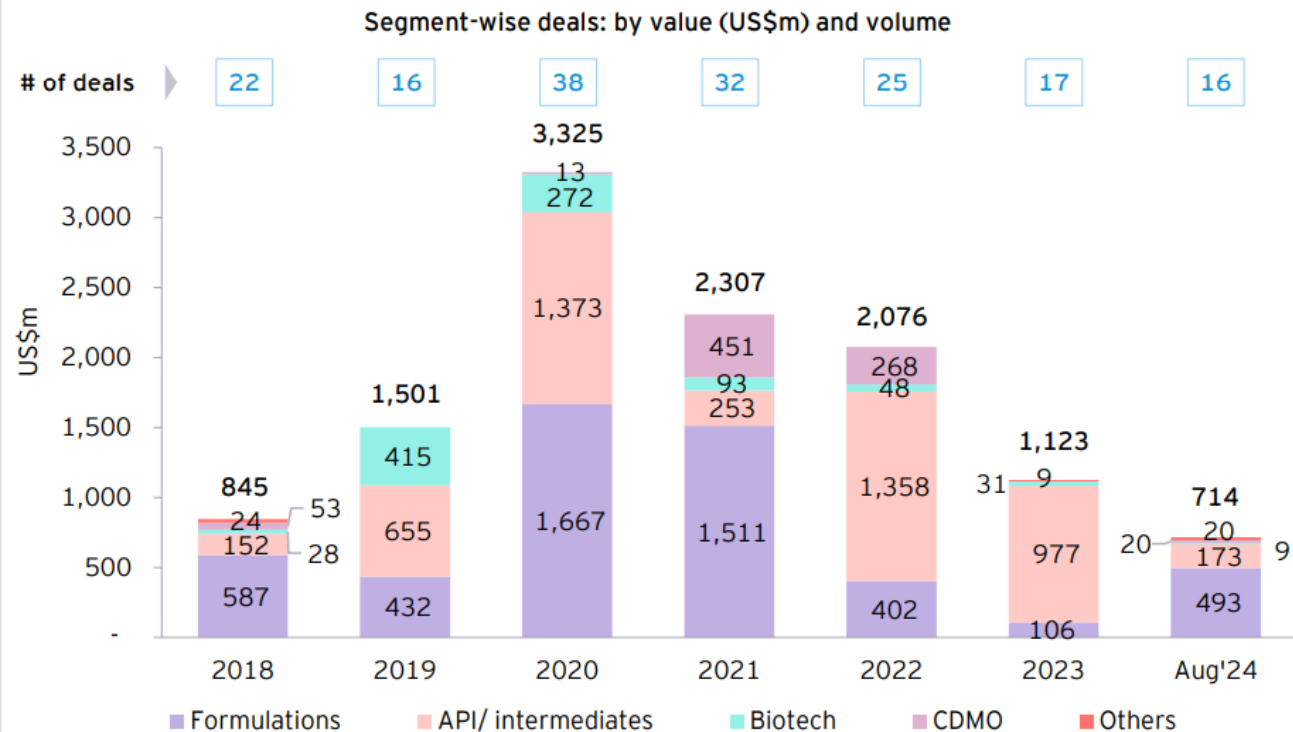
Companies under Target Segment III (Key Chemical Synthesis Based KSMs/Drug Intermediates) are as follows:

| Name of Approved Applicant         | Committed Production Capacity (in MT) | Committed Investment (in Rs. crore) | Committed Investment (in US\$ million) |
|------------------------------------|---------------------------------------|-------------------------------------|----------------------------------------|
| Saraca Laboratories Limited        | 3,000                                 | 50.0                                | 6.85                                   |
| Emmennar Pharma Private Limited    | 1,500                                 | 21.94                               | 3.01                                   |
| Hindys Lab Private Limited         | 3,000                                 | 37.60                               | 5.15                                   |
| Aarti Speciality Chemicals Limited | 4,000                                 | 77.87                               | 10.67                                  |
| Meghmani LLP                       | 13,500                                | 55.06                               | 7.55                                   |
| Sadhana Nitro Chem Limited*        | 36,000                                | 197.27                              | 27.04                                  |

- Department of Pharmaceuticals initiated the PLI scheme to promote domestic manufacturing by setting up greenfield plants with minimum domestic value addition in four separate 'Target Segments' with a **cumulative** outlay of **Rs. 6,940 crore** from FY21 to FY30.
- Under 'Target Segment I', five applications with an investment of **Rs. 3,761 crore** have been approved.
- PLI scheme for the pharmaceutical sector in India is allocated **Rs. 2,444.9 crore** (US\$ 279.77 million) for the FY26. This is a 14% increase from allocation of Rs. 2,150.5 crore (US\$ 246.08 million) in FY25.
- Companies have agreed to invest Rs. 862.01 crore in these plants, resulting in ~1,763 employment opportunities.
- The government approved a total of 19 applications totaling Rs. 4,623.01 in committed investment.

# Start-ups and PE-VC Investments

Pharmaceutical PE investment deals (2018 - YTD August 2024)



Indian pharmaceutical start-ups

## Exponential growth in the number of pharmaceutical start-ups

As of June 2024, the Department for Promotion of Industry and Internal Trade (DPIIT) has recognized 2,127 start-ups (2% of the total start-ups) in the pharmaceutical sector



The last three years have seen the setting up of 1,397 start-ups in the pharmaceutical sector

## Top five pharmaceutical PE investments: January 2018 to August 2024

| Investor-target                                  | Deal type             | Year | Deal value (US\$m) | Sub-sector        |
|--------------------------------------------------|-----------------------|------|--------------------|-------------------|
| Advent Int. Corp- Suven Pharma                   | 50% buyout            | 2022 | 770                | API/intermediates |
| New Mountain- Aurobindo Pharma, Natrol (US Unit) | 100% buyout           | 2020 | 550                | API/intermediates |
| Unison Capital- Kyowa Pharma                     | 100% buyout           | 2019 | 524                | API/intermediates |
| KKR- JB Chemicals                                | 65% buyout            | 2020 | 495                | Formulations      |
| Carlyle- Piramal Pharma                          | 20% growth investment | 2020 | 490                | Formulations      |

# Few (30+) Identified critical APIs

| API                 | Total import value (in INR crore) | Total import volume (MT) | Percentage of domestic sourcing (out of total consumption) | Average import cost (in INR) Per kg | Average domestic cost (in INR) Per kg | Largest source of import | Percentage of reliance on largest source (out of total import) |
|---------------------|-----------------------------------|--------------------------|------------------------------------------------------------|-------------------------------------|---------------------------------------|--------------------------|----------------------------------------------------------------|
| Potassium clavunate | 933                               | 1,301                    | <10%                                                       | 7,169                               | NA                                    | China                    | 59%                                                            |
| Azithromycin        | 727                               | 1,013                    | 20–40%                                                     | 7,172                               | NA                                    | China                    | 100%                                                           |
| Ceftriaxone         | 572                               | 1,229                    | <10%                                                       | 4,650                               | NA                                    | China                    | 99%                                                            |
| Meropenem           | 320                               | 58                       | 20–40%                                                     | 55,310                              | 60,143                                | China                    | 97%                                                            |
| Vitamin B12         | 304                               | 16                       | <10%                                                       | 195,895                             | NA                                    | China                    | 99%                                                            |
| Rifampicin          | 273                               | 303                      | <10%                                                       | 9,007                               | 8,910                                 | China                    | 98%                                                            |
| Acyclovir           | 222                               | 665                      | <10%                                                       | 3,337                               | 3,342                                 | China                    | 100%                                                           |
| Amoxicillin         | 222                               | 956                      | >60%                                                       | 2,317                               | 1,600                                 | China                    | 81%                                                            |
| Gabapentin          | 195                               | 697                      | >60%                                                       | 2,795                               | 2,800                                 | China                    | 99%                                                            |
| Prednisolone        | 145                               | 31                       | <10%                                                       | 46,372                              | 32,299                                | China                    | 88%                                                            |
| Clarithromycin      | 143                               | 133                      | <10%                                                       | 10,704                              | NA                                    | China                    | 100%                                                           |
| Sulbactam           | 132                               | 164                      | 20–40%                                                     | 8,071                               | NA                                    | China                    | 100%                                                           |
| Telmisartan         | 128                               | 64                       | 40–60%                                                     | 20,143                              | 12,549                                | Germany                  | 96%                                                            |
| Doxycycline         | 127                               | 225                      | <10%                                                       | 5,623                               | NA                                    | China                    | 80%                                                            |
| Betamethasone       | 126                               | 19                       | 20–40%                                                     | 65,642                              | 52,718                                | China                    | 98%                                                            |
| Levofloxacin        | 122                               | 583                      | 40–60%                                                     | 2,092                               | 2,228                                 | China                    | 100%                                                           |
| Cephalexin          | 98                                | 274                      | >60%                                                       | 3,562                               | 3,750                                 | Spain                    | 99%                                                            |

~₹4,700 Cr

| API                   | Total import value (in INR crore) |       | Percentage of domestic sourcing (out of total consumption) | Average import cost (in INR) | Average domestic cost (in INR) | Largest source of import | Percentage of reliance on largest source (out of total import) |
|-----------------------|-----------------------------------|-------|------------------------------------------------------------|------------------------------|--------------------------------|--------------------------|----------------------------------------------------------------|
| Atorvastatin          | 93                                | 34    | >60%                                                       | 27,727                       | 14,256                         | Mexico                   | 46%                                                            |
| Ciprofloxacin         | 88                                | 522   | >60%                                                       | 1,688                        | 1,715                          | China                    | 97%                                                            |
| Paracetamol           | 82                                | 2,522 | >60%                                                       | 357                          | 300                            | China                    | 91%                                                            |
| Valsartan             | 77                                | 68    | >60%                                                       | 11,335                       | 18,000                         | China                    | 90%                                                            |
| Cefoperazone          | 65                                | 75    | 20–40%                                                     | 8,675                        | NA                             | China                    | 100%                                                           |
| Losartan              | 57                                | 77    | >60%                                                       | 7,394                        | 7,500                          | China                    | 97%                                                            |
| Vitamin B1            | 55                                | 251   | <10%                                                       | 2,190                        | 1,857                          | China                    | 94%                                                            |
| Dexamethasone         | 54                                | 8     | <10%                                                       | 6,905                        | 38,833                         | China                    | 55%                                                            |
| Tetracycline          | 48                                | 194   | <10%                                                       | 2,450                        | NA                             | CHINA                    | 99%                                                            |
| Metronidazole         | 45                                | 645   | 40 - 60%                                                   | 698                          | 750                            | CHINA                    | 100%                                                           |
| Ofloxacin             | 42                                | 196   | 20 - 40%                                                   | 2,133                        | 2,339                          | CHINA                    | 100%                                                           |
| Clindamycin phosphate | 42                                | 31    | <10%                                                       | 13,263                       | NA                             | CHINA                    | 99%                                                            |
| Carbidopa             | 38                                | 14    | 20 - 40%                                                   | 28,028                       | 20,419                         | SWITZERLAND              | 71%                                                            |
| Olmesartan            | 38                                | 7     | 40 - 60%                                                   | 51,323                       | 45,000                         | NA                       | NA                                                             |
| Levodopa              | 32                                | 60    | >60%                                                       | 5,367                        | 5,198                          | JAPAN                    | 58%                                                            |
| Oxytetracycline       | 32                                | 165   | <10%                                                       | 1,946                        | NA                             | CHINA                    | 100%                                                           |
| Sulfadiazine          | 27                                | 121   | <10%                                                       | 2,232                        | 2,673                          | CHINA                    | 98%                                                            |



# Key Intermediates/ KSMs from China having high demand

| S. no. | Name of intermediate                          | Name of API synthesised               |
|--------|-----------------------------------------------|---------------------------------------|
| 1      | Cyanoacetic Acid                              | Caffeine                              |
| 2      | L-aminodiol                                   | Chloramphenicol                       |
| 3      | Azithromycin amine                            | Azithromycin                          |
| 4      | MCPSA and DCMP                                | Sulfadoxine                           |
| 5      | Cyclopropylamine                              | Ciprofloxacin                         |
| 6      | DCDA                                          | Metformin                             |
| 7      | DCFA – 2, 4-Dichloro-5-fluorobenzoyl chloride | Ciprofloxacin                         |
| 8      | 2, 3, 4, 5-tetrafluoro benzoyl chloride       | Levofloxacin ester<br>Ofloxacin ester |
| 9      | L-Alaninol                                    | Levofloxacin ester                    |
| 10     | DL-Alaninol                                   | Ofloxacin ester                       |
| 11     | 6 APA                                         | Ampicillin and Amoxicillin            |
| 12     | 7 ACA                                         | Cephalosporins                        |

Source: IDMA

PwC Reviving India's API industry

| S. no. | Name of intermediate                        | Name of API synthesised     |
|--------|---------------------------------------------|-----------------------------|
| 13     | (S,S)-2, 8 Diazabicyclo [4, 3, 0] Nonane    | Moxifloxacin                |
| 14     | 4-Cyanopyridine                             | Ethionamide<br>Protionamide |
| 15     | 4-Amino-5-(aminomethyl)-2-methyl pyrimidine | Vitamin B1 Hcl              |
| 16     | 4-Methyl-5-ethoxyloxazole                   | Vitamin B6 Hcl              |
| 17     | 2-isopropyl-4, 7-dihydro-(1,3)-dioxepin     | Vitamin B6 Hcl              |
| 18     | 2-mercaptothiazoline                        | Cysteamine Hcl              |
| 19     | 3-Cyanopyridine                             | Niacin                      |
| 20     | Beta Picoline                               | Niacin                      |
| 21     | Ethyl 3-(N,N dimethyl amino) acrylate       | Ciprofloxacin               |

**Import of Penicillin G**  
(Essential API for manufacturing  
several antibiotics from China)

Penicillin G

6000 ton

# Key Considerations – to be reviewed

## Key Consideration:

### Positives:

- Indian **API** market projected to grow to **\$22B (2029)**, CAGR **~8.5%**.
- Over **50% of APIs imported**, primarily from China.
- Has immense **export potential**.
- **Scalable and policy-supported** opportunity.
- Growing API manufacturing base, supported by the **PLI scheme and global demand**, provides a favorable environment for new entrants.
- Heavy **import dependence on China**.

### Negatives:

- Production already started for **35 of 41 APIs** under focus under PLI.
- Crowded market with top 10 players capturing ~50% of the market share in the entire pharma value chain.
- ~1,400 start-ups already in the pharma industry play.
- Significant additional production capacity already announced in the API industry and allotted under the PLI scheme.

## Recommendations:

- Identify high value > high volume API having domestic as well as export potential.
- Further deep dive is recommended to understand the gap/opportunity in the value chain.
- CAPEX requirement depends on the selection of API for manufacturing.
- Contract Manufacturing (CDMO/CRDMO) is suggested business model to further explore.
- Capital allocation strategy:
  - 50% Infra (GMP + automation),
  - 20% Technology acquisition,
  - 20% Working capital,
  - 10% Regulatory compliance.



## OUR PURPOSE

Enhance the quality of life for all stakeholders by creating a sustainable organization

## OUR VALUES



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